INCH - POUND

MIL-S-60928A(MI)
19 March 1991
SUPERSEDING
MIL-S-60928 (MI)
8 January 1968

MILITARY SPECIFICATION

SUPPORTS, CHARGE, FOR 2.75 INCH ROCKET, MARK 40 AND MARK 4

This specification is approved for use by the U.S. Army Missile Command Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the quality assurance provisions and special requirements not covered by the drawings for a type of impregnated fiberglass designated as Supports, Charge, for Rocket Motor Mark 4 and Mark 40.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification for
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1168	-	Ammunition Lot Numbering
MIL-STD-45662	-	Calibrations Systems Requirements

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

457858	Charge Support Spacer
457860	Charge Support Ring
457861	Charge Support Disc

PUBLICATIONS

EL 457858	Equipment List, Charge Support Spacer
EL 457860	Equipment List, Charge Support Ring
EL 457861	Equipment List, Charge Support Disc

(Unless otherwise indicated, requests for copies of Government drawings shall be made to the contracting officer.)

2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2) a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.1.

3.2 Material. Material shall be in accordance with the requirements specified herein, and applicable drawings and specifications.

3.3 Workmanship. All components shall be fabricated and assembled in accordance with beet practices. They shall be free of binder slugs measuring more than 0.0625 inch in diameter by 0.50 inch in length. They shall be free of layer separations or delamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inapection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.6.1, 4.6.2 and 4.8. First article inspection shall be performed by the Government

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

at a place specified by the procuring activity (see 6.2). Reference shall be made to MIL-STD-109 in order to define terms herein. The provisions of MIL-A-2550 shall apply.

4.3.1 Submission of Product. At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a) A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b) Number of units of product inspected.

c) Results obtained for all inspections performed.

d) Drawing, specification number and date, together with an identification and date of changes.

e) Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commerical specifications referenced in any of the contractual documents.

f) Number of items in the lot.

g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authorization to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.3.2 Government verification. Using the contractor's written quality assurance procedure, this detailed specification, the applicable drawings and other contractual documents aa a guide, the Government inspector shall verify at unscheduled intervals, all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.4 First article inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

4.4.1 Submission. The contractor shall submit a first article sample for evaluation as designated by the contracting officer. The first article sample shall consist of the following:

a) 50 Charge support spacers
b) 50 Charge support rings
c) 50 Charge support discs.

The first article sample shall consist of material that has been produced by the contractor or furnished by a supplier and that has been manufactured using the same production processes, procedures and equipment that will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings/certificates of conformace for materials (see 4.3.1.e). A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply so as to significantly affect product uniformity as determined by the Government.

4.5 Quality conformance inspection. Quality conformance inspection shall be performed as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

TABLE I. Quality conformance inspection.

Inspection	Requirement paragraph or drawing	Test paragraph	Method paragraph
First Article	3.1	4.4	4.8.1
Material	3.2	4.6.2	Gage/Visual
Assembly	Visual	4.6.2	Gage/Visual
Charge support spacer height load compression test	457858	4.8.1.1	4.7.1
Charge support spacer dimentional compression test	457858	4.8.1.2	4.7.2
Charge support ring height load compression test	457860	4.8.1.3	4.7.3
Charge support ring dimentional compression test	457860	4.8.1.4	4.7.4
Charge support disc height load compression test	457861	4.8.1.5	4.7.5
Workmanship	3.3	4.6.2	Visual

4.6 Quality conformance (product lot acceptance inspection).

4.6.1 Lot formation. The term “inspection lot,” as used throughout this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.6.2 Examination. The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.6.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.8.2.

4.6.2.1 Charge support spacer (see drawing 457858 and table II).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

TABLE II. Classification of defects, charge support spacer.

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:	None defined	
Minor:		
201.	Holes extending through the wall, oversize	Gage
202.	Layer separations, oversize	Gage
203.	Folds on x-x surface, oversize	Gage

4.6.2.2 Charge support ring (see drawing 457860 and table III).

TABLE III. Classification of defects, charge support ring.

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:	None defined	
Minor:		
201.	Holes extending through the wall, oversize	Gage
202.	Layer separations, oversize	Gage
203.	Folds on x-x surface, oversize	Gage

4.6.2.3 Charge support disc (see drawing 456861 and table IV).

TABLE IV. Classification of defects, charge support disc.

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:		
101.	Folds on x-x surface, oversize	Gage
Minor:		
201.	Disc diameter	Gage

4.7 Test Methods and procedures.

4.7.1 Charge support spacer height-load compression test (see 4.8.1.1). The charge support spacer shall be cycled five times from free height to a

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

height of 0.2 ± .02 inches. When compressed to a height of 0.28 inch, the spacers must meet force requirements. When compressed to a height of 0.2 inches, the spacer must meet force requirements. Load tests shall be conducted by compressing between flat parallel metal plates.

4.7.2 Charge support spacer compression test (see 4.8.1.2) . The charge support spacer shall be compressed to 0.2 inches. Load tests shall be conducted by compressing between flat parallel metal plates.

4.7.3 Charge support ring height-load compression test (see 4.8.1.3) . The charge support ring shall be cycled five times from a free height to a height of 0.76 ± .02 inches. When compressed to the required height the ring must meet force requirements. Load tests shall be conducted by compressing between flat parallel metal surfaces.

4.7.4 Charge support ring corepression test (see 4.8.1.4) . The charge support ring shall be compressed to 0.76 inches. Load tests shall be conducted by compressing between flat parallel metal plates.

4.7.5 Charge support disc height-load compression test (see 4.8.1.5) . The charge support disc shall be cycled five times from a free height to a height of 0.13 ± .01 inches. when compressed to a height of 0.16 inches the disc must meet force requirements. When compressed to a height of 0.13 inches, the disc must meet force requirements. Load test shall be applied by means of hollow cylinder, 2.562 inches outer diameter (O.D.) x 1.750 inches inner diameter (I.D.) with the disc resting on a flat metal plate.

4.8 Testing and production.

4.8.1 First article (see 4.4). The first article sample shall be examined 100 percent for all of the characteristics in 4.6.2, and then subjected to the tests listed below. Inspection and test equipment shall conform to the requirements of 4.8.2. Rejection of the first article sample shall be in accordance with 4.4.1.

4.8.1.1 Charge support spacer height-load compression test (see drawing 457858. Major defect. Fifty samples from each lot of charge support spacers shall be tested as specified in 4.7.1 using equipment in accordance with 4.8.2. A charge support spacer that does not meet the height-load requirements as specified on the drawing when compressed as required shall be considered a defective item. One or more defective items shall reject the lot.

4.8.1.2 Charge support spacer dimensional compression test (see drawing 457858). Minor defect. Fifty samples from each lot of charge support spacers shall be tested as specified in 4.7.2 using equipment in accordance with 4.8.2. A charge support spacer that does not meet the dimensional requirements as specified on the drawing when compressed as required shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

considered a defective item. One or more defective items shall reject the lot.

4.8.1.3 Charge support ring height-load compression test {see drawing 457860). Major defect. Fifty samples from each lot of charge support rings shall be tested as specified in 4.7.3 using equipment in accordance with 4.8.2. A charge support ring that does not meet the height-load requirements as specified on the drawing when compressed as required shall be considered a defective item. One or more defective items shall reject the lot.

4.8.1.4 Charge support ring dimensional compression test (see drawing 457860). Minor defect. Fifty samples from each lot of charge support rings shall be tested as specified in 4.7.4 using equipment in accordance with 4.0.2. A charge support ring that does not meet the dimensional requirements as specified on the drawing when compressed as required shall be considered a defective item. One or more defective items shall reject the lot.

4.8.1.5 Charge support disc height-load compression test (see drawing 457861) . Major defect. Fifty sample from each lot of charge support discs shall be tested as specified in 4.7.5 using equipment in accordance with 4.8.2. A charge support disc that does not meet the height-load requirements as specified on the drawing when compressed as required shall be considered a defective item One or more defective items shall reject the lot.

4.8.2 Inspection equipment. Equipment List numbers EL-457858, EL-457860, and EL-457861 identify the inspection equipment required to perform the examinations and tests required by 4.6.2 and 4.8 of this specification-

4.8.2.1 Calibration. The contractor shall have available a set of master gages, standards and appropriate instruments for regularly scheduled calibrations of his acceptance inspection equipment. Records of such regularly scheduled calibrations shall be maintained by the contractor and made available for review by the Government. The calibration of gages, standards and instruments shall be in accordance with MIL-STD-45662.

4.8.2.2 Supply and maintenance. Supply and maintenance of the equipment listed on the EL shall be in accordance with MIL-I-45607.

5. PACKAGING

5.1 Preservation, packing and marking. The fiberglass charge-supports shall be packed correctly and legibly marked in accordance wit MIL-STD-129. Charge supports shall not be mixed in the individual shipping box.

5.2 Marking. The exterior of each shipping container shall be correctly and legibly marked in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The supports covered by this specification are intended for use in Mark 40 and Mark 4 2.75 inch rocket charge assemblies.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification and applicable drawings

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. Data cards. Data cards shall be prepared for each lot in accordance with MIL-STD-1167

d. Whether a first article sample is required (see 3.1 and 4.4)

e. Applicable levels of preservation (see 5.1)

f. Location of inspection facilities (See 4.3).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (keyword) listing.

Charge, explosive
Explosives
Rocket assemblies
Support mechanism

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

6.5 Metrication. Metric equivalents, in accordance with FED-STD-376, are acceptable for use in this specification.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Preparing Activity:
Army - MI
Project No. 1340 - A760

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-60928A (MI)	19 March 1991

3. DOCUMENT TITLE
SUPPORTS, CHARGE, FOR 2.75 INCH ROCKET, MARK 40 AND MARK 4

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

NAME COMMANDER U.S. ARMY MISSILE COMMAND	b. TELEPHONE (Include Area Code) (1) Commercial: (205) 876-6980 (2) AUTOVON: 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD FORM 1426

Source: https://assist.dla.mil -- Downloaded: 20260-23T01:06Z
Check the source to verify that this is the current version before use.